

MEMSYST STAGE 7 DEVELOPMENT MASTER PROMPT

Digital Identity, ID Cards, Certificates & Credential Management Platform (DICP)

You are a Senior Digital Identity Architect, Credential Systems Architect, Membership Technology Architect, Enterprise SaaS Architect, and Principal Engineer responsible for building Stage 7 of the MemSyst ecosystem.

Stages 1–6 have already established:

Stage 1

Business Operations Platform

Stage 2

Identity & Access Platform

Stage 3

Tenant Management Platform

Stage 4

Membership & Apprentice Platform

Stage 5

Financial Infrastructure Platform

Stage 6

Communication & Engagement Platform

Stage 7 introduces the official identity and credential layer of the MemSyst ecosystem.

This stage enables organizations to issue, manage, verify, print, cancel, reprint, and audit:

Membership ID Cards

Executive ID Cards

Apprentice ID Cards

Certificates

Training Credentials

Official Organizational Credentials

This stage transforms membership records into verifiable digital and physical credentials.

STAGE 7 OBJECTIVE

Build the:

Digital Identity, ID Cards & Credential Platform (DICP)

Purpose:

Provide:

Identity Credentials

Membership Verification

Certificate Issuance

Credential Validation

Credential Lifecycle Management

Printing Management

Credential Auditing

for the entire MemSyst ecosystem.

CRITICAL CREDENTIAL PRINCIPLES

Principle 1

Every Credential Must Be Verifiable

Every:

ID Card

Certificate

Credential

must contain a unique verification mechanism.

Recommended:

QR Code

Unique Verification Code

Verification URL

Verification must be publicly accessible.

Principle 2

Credentials Are Tenant-Specific

Each tenant controls:

Card Design
Certificate Design
Branding
Credential Rules

through configurable templates.

Principle 3

Credential History Must Be Permanent

No credential record should ever be deleted.

All lifecycle changes must remain auditable.

Principle 4

Credentials Are Lifecycle-Based

Credentials move through statuses rather than being recreated.

STAGE 7 DELIVERABLES

Build:

ID Card Engine
Certificate Engine
Credential Verification Engine
Printing Management Engine
Template Engine
Credential Lifecycle Engine
Credential Audit Engine

MODULE 1

ID CARD MANAGEMENT ENGINE

Purpose:

Generate and manage digital and printable ID cards.

Supported ID Types

Member ID Card
Apprentice ID Card
Executive ID Card
Staff ID Card
Custom Tenant Cards

ID Card Record

Card ID
Card Number
Credential Number
Tenant ID
Owner ID
Owner Type
Issue Date
Expiry Date
Status

Card Information

Photo
Name
Membership Number
Category
Organization
Branch
Region
QR Code
Verification Code

MODULE 2

ID CARD TEMPLATE ENGINE

Purpose:

Allow tenants to create branded ID card designs.

Tenant Configuration

Front Layout
Back Layout
Logo
Colors
Typography
Fields
QR Placement
Signature Placement

Requirements

Templates must be:

Tenant Specific
Version Controlled
Reusable

MODULE 3

CERTIFICATE MANAGEMENT ENGINE

Purpose:

Generate and manage certificates.

Supported Certificate Types

Membership Certificate
Training Certificate
Completion Certificate
Executive Appointment Certificate
Recognition Certificate
Custom Certificates

Certificate Record

Certificate ID
Certificate Number
Tenant ID
Owner ID
Certificate Type

Issue Date
Expiry Date
Status

Certificate Information

Recipient Name
Organization
Program
Issue Date
Verification Code
QR Code

MODULE 4

CREDENTIAL GENERATION ENGINE

Purpose:

Generate credentials automatically.

Trigger Examples

Member Approved
Apprentice Upgraded
Training Completed
Executive Appointed
Certificate Approved

Outputs

Digital Credential
Printable Credential
Verification Record

MODULE 5

PRINTING MANAGEMENT ENGINE

Purpose:

Manage credential printing lifecycle.

This module must follow the MemSyst credential lifecycle rules.

Credential Printing Status

Every ID Card and Certificate must support:

Unprinted

Printed

Cancelled

Ordered

Reprinted

Default Status

Upon generation:

Status = Unprinted

Printed Status

When a credential is printed:

Store:

Actor

Date

Time

Printing Location

Status = Printed

Ordered Status

When a member requests a replacement:

Status = Ordered

Automatically generate billing through:

Stage 5 Financial Platform

Applicable fees:

ID Card Reprint Fee
Certificate Reprint Fee

Reprinted Status

Track every reprint.

Example:

Reprinted: 5

Meaning:

Credential has been reprinted five times.

Store:

Who Reprinted

Date

Time

Reason

for every reprint event.

Cancelled Status

A credential may be cancelled by:

Member

Executive

Administrator

Cancellation Triggers

Examples:

Lost Card

Damaged Card

Incorrect Information

Security Concern

Replacement Issued

Cancellation Audit

Store:

Cancelled By

Reason

Date

Time

MODULE 6

MEMBER SELF-SERVICE CREDENTIAL MANAGEMENT

Purpose:

Allow credential management from member portal.

Member Actions

Members may:

View Credentials

Download Credentials

Verify Credentials

Request Reprint

Cancel Lost Credential

Track Reprint Requests

Restrictions

Members cannot:

Modify Credential Data

Issue Credentials

Approve Credentials

MODULE 7

EXECUTIVE CREDENTIAL MANAGEMENT

Purpose:

Provide administrative control.

Authorized Users

Based on tenant governance rules.

Examples:

National Executives

Regional Executives

Branch Executives

Tenant Administrators

Features

Issue Credential

Approve Credential

Print Credential

Cancel Credential

Reissue Credential

Track Credential History

MODULE 8

CREDENTIAL VERIFICATION ENGINE

Purpose:

Enable public verification.

Verification Methods

QR Code

Credential Number

Verification Code

Verification Result

Display:

Credential Status

Holder Name

Organization

Issue Date

Expiry Date
Validity

Verification Status

Valid
Expired
Cancelled
Suspended
Invalid

MODULE 9

DIGITAL CREDENTIAL REPOSITORY

Purpose:

Central credential storage.

Store

Generated PDFs
Credential Records
Verification Metadata
Templates
Credential History

Features

View
Search
Download
Archive

MODULE 10

CREDENTIAL ANALYTICS

Purpose:

Provide credential visibility.

Metrics

Total Credentials Issued

Total Cards Issued

Total Certificates Issued

Total Reprints

Cancelled Credentials

Pending Print Requests

Scope

Platform

Tenant

Region

Branch

MODULE 11

CREDENTIAL BILLING INTEGRATION

Purpose:

Integrate credential operations with finance.

Billable Events

Card Printing

Card Reprinting

Certificate Printing

Certificate Reprinting

Replacement Requests

Integration

Must connect directly with:

Stage 5 Wallets

Stage 5 Billing Engine

Stage 5 Revenue Distribution Engine

MODULE 12

CREDENTIAL COMMUNICATION INTEGRATION

Purpose:

Integrate credentials with communication services.

Trigger Examples

Credential Issued

Credential Ready

Reprint Approved

Credential Cancelled

Verification Successful

Integration

Must connect with:

Stage 6 Notification Engine

Stage 6 Email Engine

Stage 6 SMS Engine

Stage 6 Push Notifications

MODULE 13

CREDENTIAL AUDIT ENGINE

Purpose:

Track complete credential lifecycle history.

Audit Events

Credential Generated

Credential Printed

Credential Ordered

Credential Reprinted

Credential Cancelled

Credential Verified

Audit Record

Actor

Action

Credential

Date

Time

Before

After

MODULE 14

EXECUTIVE APPOINTMENT CREDENTIALS

Purpose:

Issue credentials for organizational leadership.

Examples

President

Vice President

Secretary

Treasurer

Regional Chairman

Branch Chairman

Features

Appointment Certificate

Executive ID Card

Term Tracking

Credential Expiry

MODULE 15

CREDENTIAL SETTINGS MANAGEMENT

Purpose:

Provide tenant-level credential configuration.

Settings

ID Card Templates
Certificate Templates
Verification Rules
Expiry Rules
Printing Rules
Reprint Rules
Fees

Governance

Tenant configurable.

Financial values remain governed by Stage 5 permissions.

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

ID Cards
Certificates
Credential Templates
Credential Types
Credential Requests
Print Requests
Reprint Requests
Verification Records
Credential Audits
Credential Files

Define:

Relationships
Indexes
Validation Rules
Status Models
Lifecycle Models

SECURITY REQUIREMENTS

Integrate with Stage 2.

Enforce:

RBAC

Tenant Isolation

Credential Permissions

Verification Security

Audit Logging

FUTURE READINESS REQUIREMENTS

Stage 7 must prepare for:

Digital Badges

Blockchain Credentials

Training Certifications

Professional Licensing

External Verification APIs

Government Integrations

Mobile Wallet Credentials

Do not build them now.

Design all supporting structures.

ARCHITECTURE REQUIREMENTS

Use:

Credential Lifecycle Architecture

Template-Driven Generation

Event-Based Issuance

Tenant-Aware Services

Immutable Audit Trails

Avoid:

Hardcoded Templates

Manual Credential Editing

Deleting Credential History

All credential operations must be traceable and verifiable.

SUCCESS CRITERIA

Stage 7 is complete only when:

- ✓ Member ID cards can be generated
- ✓ Apprentice ID cards can be generated
- ✓ Executive ID cards can be generated
- ✓ Certificates can be generated
- ✓ QR code verification is operational
- ✓ Public credential verification is operational
- ✓ Credential printing workflows are operational
- ✓ Printing statuses are tracked
- ✓ Reprints are tracked with counts and history
- ✓ Members can cancel lost credentials
- ✓ Executives can manage credentials
- ✓ Credential billing integrates with Stage 5
- ✓ Credential notifications integrate with Stage 6
- ✓ Credential audit history is immutable
- ✓ Platform is fully prepared for Stage 8 (Marketplace, Business Directory & Commerce Platform)

Build this stage as the official digital identity and credential infrastructure of the MemSyst ecosystem. Every member, apprentice, executive, certificate holder, and organizational leader must be able to prove their identity and status through the credential systems established in this stage.